

Saturday Test — No. 7

Take your time. Show your work where it helps.

Name: _____

Date: _____

Time: 15 min

Section A — Numbers and Patterns

1. Work these out, then look at your answers.

odd + odd: $5 + 7 = \underline{\quad}$ $3 + 9 = \underline{\quad}$

even + even: $4 + 8 = \underline{\quad}$ $6 + 2 = \underline{\quad}$

odd + even: $5 + 4 = \underline{\quad}$ $7 + 6 = \underline{\quad}$

When do you get an *odd* answer? When do you get an *even* one?

2. Two whole numbers add up to 15. Can both of them be even? Can both of them be odd?

3. Without adding them one at a time, decide whether $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$ is odd or even.

4. Can the numbers 1, 2, 3, 4, 5, 6 be split into two groups that have the same total?

Section B — Counting Carefully

5. Seven cups sit upside-down on a table. Each move, you must turn over exactly two cups. Can you ever get all seven cups right-side up?
6. A grasshopper sits at 0 on a number line. It makes exactly nine jumps. Each jump moves it one step left or one step right. Can it land back on 0 after the ninth jump?

Section C — Reasoning

7. Twenty-three children want to split into teams of exactly two for a game. It doesn't come out evenly. Their teacher offers to join in. Does adding the teacher fix it — or does she just need a partner too?

8. Nine children sit around a round table. You want to give each child a hat, red or blue, so that no child sits next to someone wearing the same color. Can it be done?

Answer Key — for parents

1. **odd + odd = even, even + even = even, odd + even = odd.** (12, 12; 12, 8; 9, 13.) A sum is odd *only* when exactly one of the two numbers is odd. This is the tool for the whole test.
2. **No to both.** Two evens add to an even number, and two odds also add to an even number — but 15 is odd. So one number must be odd and the other even.
3. **Odd.** The even numbers (2,4,6,8,10) add to an even total; the odd numbers are 1,3,5,7,9 — five of them. An odd count of odd numbers makes the whole sum odd. (It is 55.)
4. **No.** The six numbers add to 21, which is odd. Two equal groups would need each to total $21 \div 2$, and you can't split an odd number into two equal whole-number halves.
5. **No.** Start with 0 cups up — an even number. Flipping exactly two cups changes the count of upright cups by 2, 0, or -2 , so it stays even forever. 7 is odd, so it's out of reach. (The number of upright cups is the hidden thing that never turns odd.)
6. **No.** To return to 0 it needs the same number of left jumps as right jumps — so an even number of jumps in total. Nine is odd. (After an odd number of ± 1 steps the grasshopper is always on an odd number, never on 0.)
7. **It works.** 23 is odd, so the children alone always leave one person without a partner. Adding the teacher makes 24, which is even, so they form 12 full teams of two. The trap is thinking “she needs a partner too,” but she's already counted in the 24.
8. **No.** Around the table the colors must alternate red, blue, red, blue, ... With an even number of seats that works, but 9 is odd: follow the colors all the way around and the ninth child ends up next to the first child with the *same* color. (This is the same odd/even wall as the others.)